Piano learning app design

A beginner web app for reading notation, understanding harmony and practising with MIDI feedback.

Read notation

See one or more written notes, play them, then learn from the difference.

Understand harmony

Connect chord symbols, staff notation, keyboard shapes and sound.

Build fluency

Practise scales and arpeggios in any key, with clear, measured goals.

THE LEARNING LOOP

01

Understand

One small musical idea

02

Try with help

Hear, see and play

03

Recall

Remove the hints

04

Apply and review

Use it in a new example

NAVIGATION

Learn | **Daily trainer** | **Practise** | **Explore** | **Progress** | **Devices**

Connection scope

USB and Bluetooth MIDI are both required. Desktop, optional app piano and no mandatory account remain draft assumptions.

A small step at a time

Reading and harmony grow together. Musical application gives each drill a purpose.

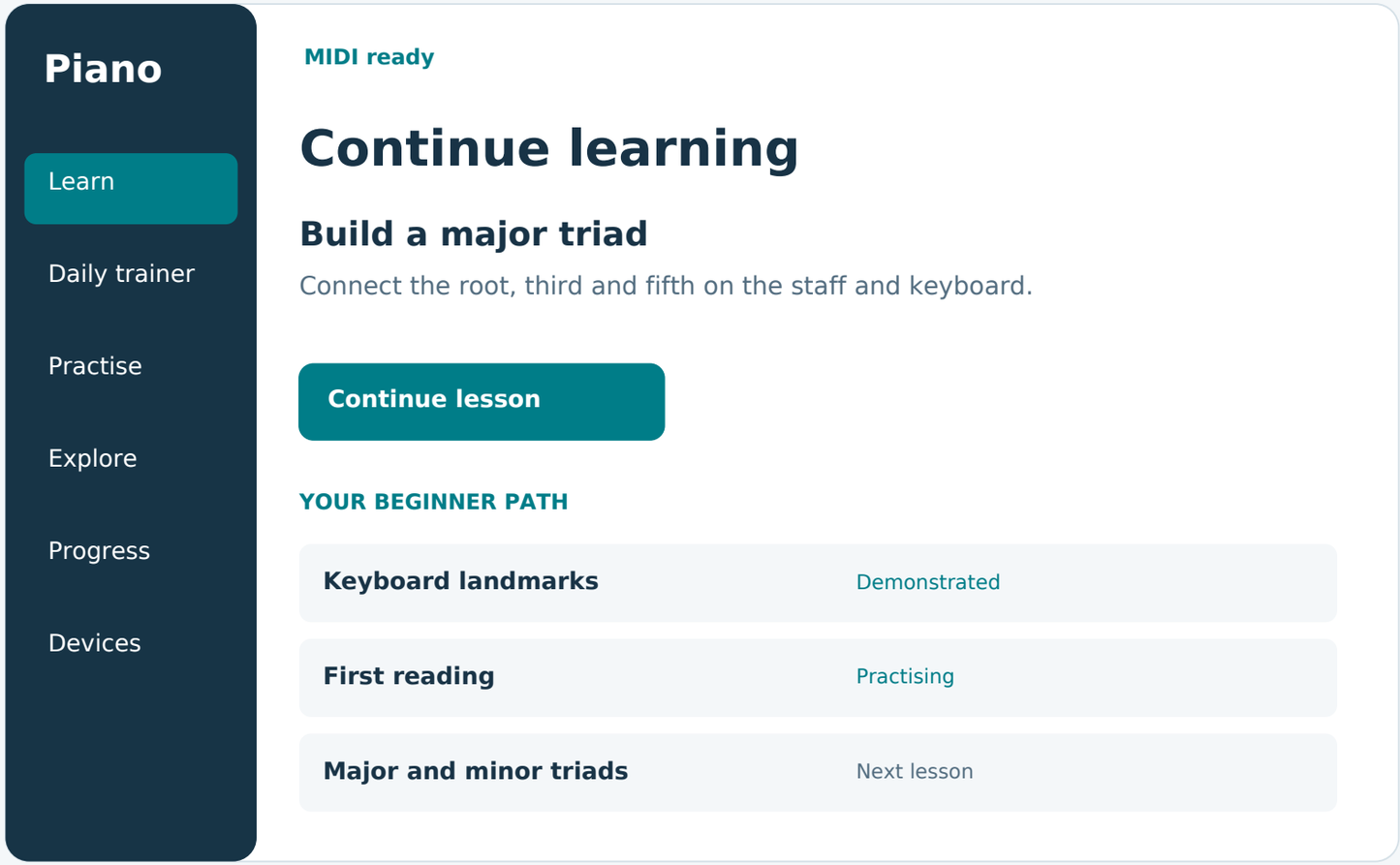
Stage	Understand	Play and observe
1 Keyboard and pulse	Note letters, black-key groups, middle C	Find notes; repeat a note with a steady beat.
2 First reading	Treble and bass landmarks, steps and skips	Read a small range, then short one-hand phrases.
3 Intervals and scales	Semitones, tones and the major-scale pattern	Play intervals and five-note patterns; add new keys.
4 Triads and inversions	Root, third and fifth; major versus minor	Build, alter and read chords; identify the bass note.
5 Harmony in motion	Scale degrees, common tones and cadences	Play C-F-G-C, broken chords and short arpeggios.
6 Reading and application	Grand staff, coordination and key signatures	Read unfamiliar studies; transpose a learned idea.

Introduce keys gradually while keeping all keys available

Lessons can start in C, then G and F. Key-based drills support all 12 tonic roots from the first release.

Open on the next useful lesson

A compact learning path supports beginners while keeping focused practice easy to reach.



ONE CONCEPT

Short explanations link the theory, staff and keyboard. Demonstrations can be replayed or slowed.

A LESSON SEQUENCE

Understand -> observe -> try with help -> recall -> apply. A typical lesson takes 5-10 minutes.

LEARNER CONTROL

Placement is optional. Open Daily trainer for a short routine, or browse focused drills. Choose either start tab.

Evidence behind progress

Hints and demonstrations support learning. Independent checks remove answer cues and use fresh examples.

A daily trainer you can just start

Open the tab, press Start and follow a prepared routine. Settings are remembered; adjustments stay optional.

DAILY TRAINER

0 of 4 tasks done

Your 10-minute routine

1 Settle in

Five-note pattern / hands separately / 2 min

2 Finger-pattern control

Neighbor pairs / Level 2 / G major / 3 min

3 Steady speed

60 BPM / quarter notes / reset breaks / 3 min

4 Apply in a key

G-major arpeggio / Level 1 / 2 min

Start today's session

Adjust routine

READY WHEN YOU ARE

The routine uses saved skill levels, keys and tempos. First use gets simple beginner defaults; no daily setup is required.

ONE CONTINUOUS SESSION

Demonstrations, count-ins, short attempts and reset breaks lead into the next task automatically. Pause, Slower and Skip stay visible.

LEVELS AND SPEED

Control, pulse and coordination progress separately. Tempo increases need qualifying attempts; MIDI cannot verify finger use.

Start once then follow the trainer

Default to 10 minutes. Offer Resume when unfinished and a clear Done state at the end. Missed days do not erase progress.

Read the staff then play the notes

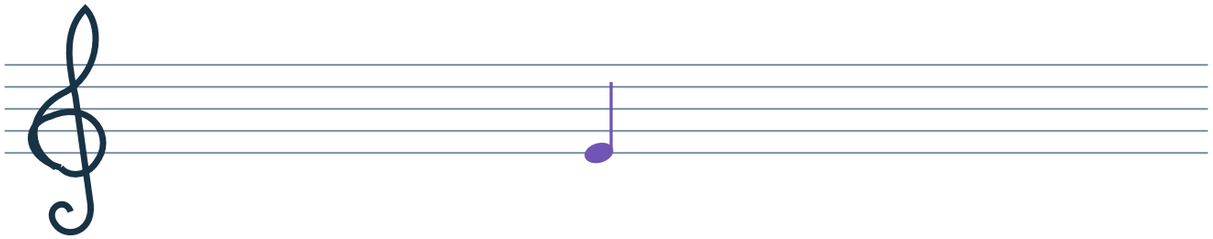
Start untimed. Add chords, short phrases and a steady musical timeline as confidence grows.

GUIDED

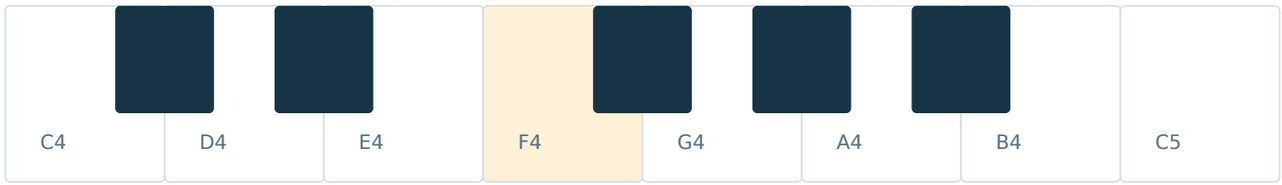
Treble landmarks

Item 4 of 10

Play the written note



You played F4. Try one step lower.



Detected key

Hint

Pause

ONE OR MORE NOTES

Single notes progress to intervals, exact written chords and short phrases. Octave is part of the answer.

SUPPORT FADES

Target-key highlights and note names help during teaching. Reading checks hide them; revealing a key records assistance.

USEFUL CORRECTIONS

Distinguish a different pitch from the correct note name in the wrong octave. Holding one key cannot answer several prompts.

Notation and pitch are separate

C-sharp and D-flat share a key. Ask for a written-name answer when testing spelling; MIDI alone cannot distinguish intent.

Make harmony visible and audible

Compare chord tones, hear the change, then rebuild the harmony from memory.

C major

Root + major third + perfect fifth



C minor

Lower the third from E to E-flat



TEACH THROUGH CHANGE

Build a chord, change its third, find a missing tone, then compare inversions.

EXPLAIN THE BASS

In E-G-C, the root is C and the lowest note is E. Teach those as different ideas.

THE ROOT STAYS C

C - E - G

Root position

E - G - C

First inversion

G - C - E

Second inversion

APPLY THE IDEA

Introduce I-IV-V-I beside C-F-G-C. Add common-tone and nearby-voicing tasks after the basic shapes.

A named chord can allow several voicings. A written chord requires its exact notes.

Free play is exploratory

Ambiguous or incomplete note sets show notes or possible names. They do not receive an invented correctness score.

Practise arpeggios in any key

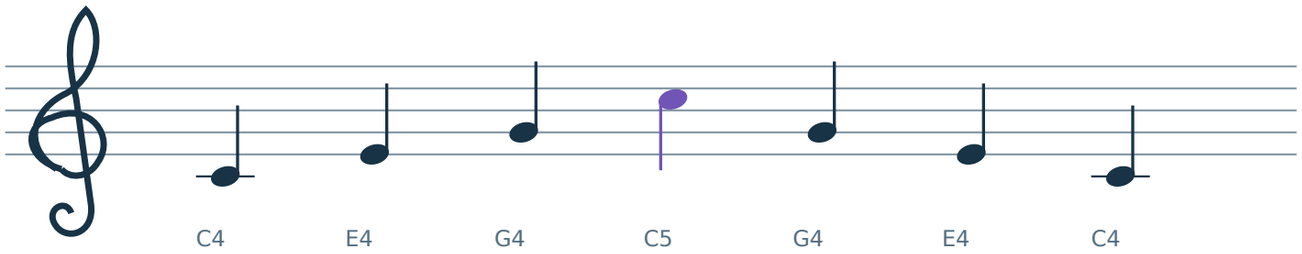
Select the key and pattern. Notation, demonstrations and MIDI targets change together.

ARPEGGIO

Key: C major

Change key

C4 - E4 - G4 - C5 - G4 - E4 - C4



Seven attacks. The top C is played once.

Mode	Range	Repetitions
Guided	**C4 to C5**	**3**

THE SAME IDEA IN A NEW KEY

C major: C-E-G-C. G major: G-B-D-G. D major: D-F-sharp-A-D. Minor tonic arpeggios use the minor third.

ONE KEY OR SELECTED KEYS

Stay in one key, or rotate through selected keys in a chosen, fifths or shuffled order. Reset between keys.

SEPARATE EVIDENCE

Record accuracy and timing per key, tonality and tempo. A success in C does not imply fluency in every key.

Keep the notation musically correct

All 12 roots are available. Key signature, spelling, octave range and expected MIDI notes must agree.

Choose a drill with a clear goal

Templates cover the beginner tasks. Setup shows the exact range and the rules before practice begins.

PRACTICE CATALOGUE

Note reading

Clef, range and accidentals

Harmony

Quality, inversion and key

Patterns

Scale, arpeggio or broken chord

Rhythm

Pulse, rests and duration

Short studies

Reading in musical context

Listening

Intervals and chord quality

Tonic arpeggio

Setting	Selected
Key and tonality	G major / change
Pattern	Up and down / right hand
Range	G3 to G4 / fits your keyboard
Mode	Guided / 3 repetitions
Keys in session	G only / add selected keys

Save preset | **Start practice**

Make changes explicit

An exercise that exceeds the keyboard range offers a narrower pattern or another task. Written notes are never silently transposed.

Two modes with different promises

Pitch discovery needs time. Rhythm practice needs a reference that keeps moving.

Guided practice

Wait for the correct response. Explain one mismatch at a time. Store retries and hints. No speed penalty.

Timed practice

Count in, then keep the score timeline moving. Match pitch and onset; review detailed corrections after the phrase.

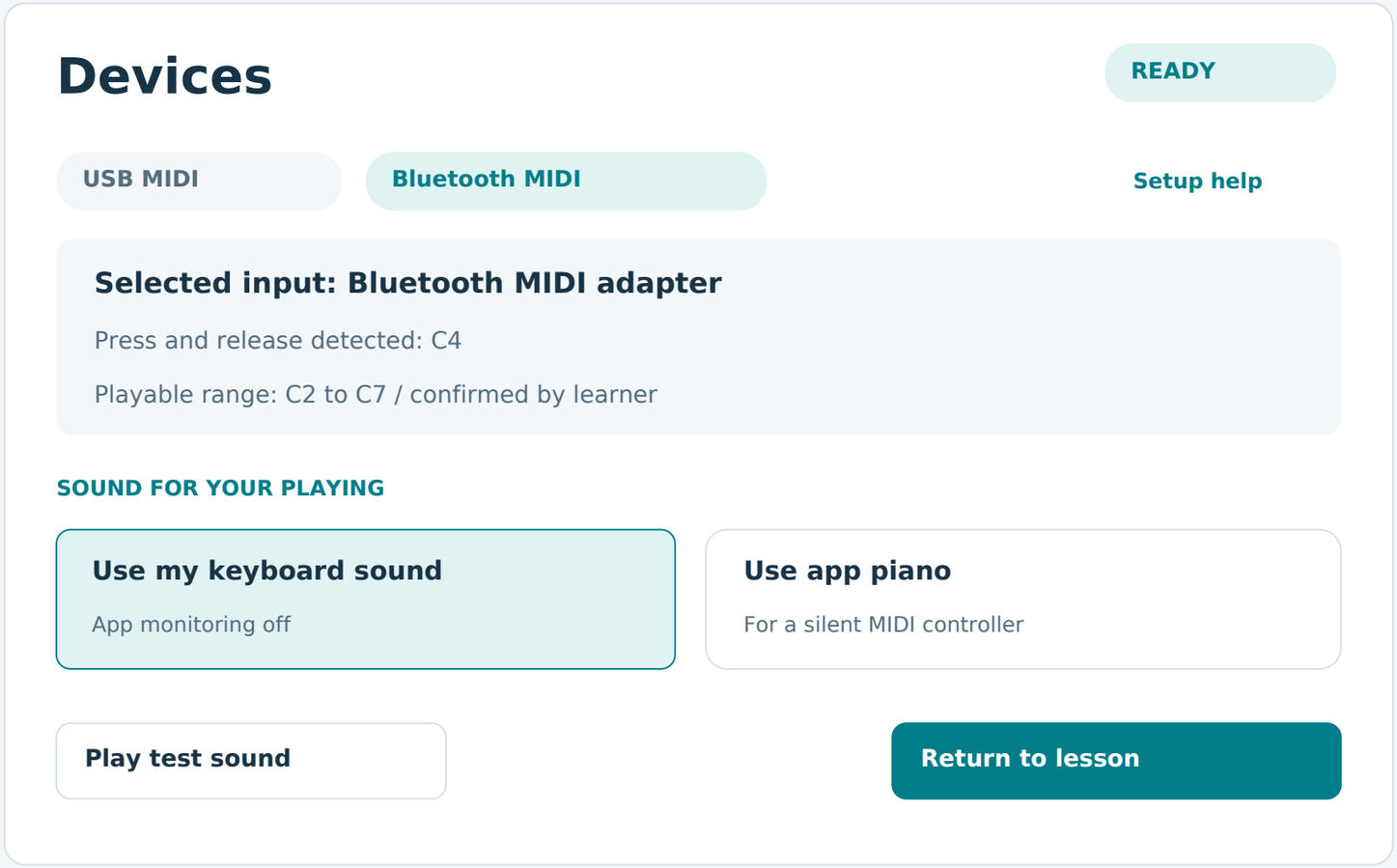
Task	Acceptance rule
Single written note	Fresh exact pitch and octave. Release before repeating the same key.
Written chord	Exact held set; no prohibited extra notes. Guided settle time starts at 120 ms.
Chord name or inversion	Declared chord tones, allowed range, bass-note rule and doubling policy.
Timed sequence	One attack matches at most one target. A wrong note does not shift later targets.

Show separate evidence

First-try accuracy, eventual completion, hints, missed notes, extras and timing stay distinct. Thresholds are provisional.

Connect through USB or Bluetooth MIDI

Both connection types are in the first release. Pairing, receiving notes and hearing sound are separate states.



ADAPTERS INCLUDED

Support external BLE-MIDI adapters and built-in Bluetooth MIDI. Test the exact adapter, OS, browser and route.

PAIR THEN SELECT MIDI

Guide system MIDI setup or a tested wireless receiver. Return to the app and select the exposed MIDI input. [6][7]

TEST AFTER RECONNECT

Pairing alone is not Ready. Verify notes and releases again; select one input if USB and Bluetooth both appear.

MIDI connection and sound are separate

Bluetooth MIDI carries performance events. It is distinct from Bluetooth audio. A silent controller can still use the app piano.

Measure what the input can tell you

Keep useful evidence without claiming to observe technique that MIDI does not report.

Observable	Use in feedback	Boundary
Note attacks and releases	Pitch, order, register and observed key overlap	Cannot establish finger identity or hand posture.
Event timestamps	Onset error against the fixed score timeline	Preserve clock mapping; flag unreliable timing.
Velocity and pedal messages	Optional velocity consistency; separate sustain	Velocity is not acoustic loudness; devices differ.

PROPOSED TIMING RULE

Tolerance = the smaller of 200 ms or one-quarter of the inter-onset interval. At 60 BPM with quarter notes: +/-200 ms. Store the actual profile with the attempt.

RECOVERY FLOW

Pause -> clear stale keys -> release keys and pedal -> test input -> count in for timed work. Restart the interrupted item or bar; retain completed segments.

USB and Bluetooth share musical rules

Test timing by route. Do not guess a Bluetooth offset or mistake uneven playing for a bad link. Recheck timing context after reconnection.

Finish with evidence and a useful choice

Review the specific task. Make the next step optional and explain why it may help.

Treble landmarks complete

10 items / Guided / Saved

8 of 10	2	1
Correct first try	Corrected	Used a key hint

Try these two notes again

E4 and F4 caused the two corrections. Practise a fresh short set around them.

Retry these items

Done

HONEST TOTALS

All 10 items were eventually completed. Only 8 were correct on the first attempt; one item used an answer hint.

SPECIFIC SKILLS

Track reading and harmony alongside daily movement levels and qualified tempos, separately for each key and pattern.

FRESH CHECKS

A proposed milestone uses two fresh 9-of-10 unassisted checks on different sessions. It is a heuristic, not lasting-mastery proof.

Protect the meaning of a result

Guided success is not a timed-fluency claim. Replaying a known phrase is useful practice but is not fresh sight-reading.

Keep targets performances and progress separate

Snapshot what the learner was asked to do so a saved result keeps its meaning after content changes.

Content and preset

Skill, spelling, pattern and rules

Resolved attempt

Exact notes, mode and tolerances

Events and outcomes

What happened, with assistance

Record	Preserve
Definition and routine	Content revision, parameters, dated daily tasks, skill levels and linked attempts.
Attempt snapshot	Targets, key/tonality, key order, evaluator, tempo, range, input route and timing context.
Performance evidence	Note/controller events, timestamp source, hints, connection changes and per-target results.
Progress view	Evidence, per-skill levels, qualified tempos and rule version; rebuildable from attempts.

Local autosave and portable backup

JSON restores the complete record. CSV supports analysis. Browser storage can be cleared and is not cross-device synchronization.

Build one complete learning journey first

Prove the device and evaluation behavior before expanding the amount of curriculum.

Technical proof first

Test USB and Bluetooth adapters: notes, releases, chords, pedal, fast arpeggios, link loss, reconnect and timing capture.

First complete slice

- Connect and test one note.
- Learn a keyboard landmark.
- Read notes in guided mode.
- Play a timed two-bar pattern.
- Review, export and restore.

Beginner first release

- USB and Bluetooth MIDI with app sound.
- Daily trainer with levels and speed goals.
- Treble/bass reading, triads and inversions.
- Scales and arpeggios in all 12 roots.
- Selected-key practice and per-key evidence.

Later expansion

- Deeper key lessons and minor scales.
- Richer listening and two-hand studies.
- General exercise editor and imports.
- Teacher accounts and cloud sync.
- Mobile and more adapter combinations.

Keep early scope focused

No microphone recognition, camera technique scoring, automatic fingering verification, social feed or leaderboard is required.

Three answers would sharpen the design

The draft can move forward with its stated assumptions. These choices would most change the product.

1 First hardware setup

Which keyboard, Bluetooth MIDI adapter and computer/browser? Does the keyboard make its own sound?

2 Musical applications

Should short studies lean toward pop and chord accompaniment, classical reading, or a balance?

3 Intended audience

Is this primarily a personal practice tool, or a product for other beginners as well?

Working default	Decision before implementation
USB and Bluetooth MIDI required	Confirm the first desktop, adapter and browser matrix.
Authored beginner content	Choose lesson volume, keys, ranges and educator review.
Local data without an account	Confirm portability, recovery, retention and any later sync need.
Configurable timing and progress rules	Test tolerances and milestone language with beginners.

Design status

All screens, readings and progress values are illustrative. The complete Markdown specification defines the intended behavior.

Verify the promises the learner relies on

Use real devices and replayable event cases to check musical correctness and recovery.

- A new learner can use USB or a Bluetooth MIDI adapter and complete a note test.
- Daily trainer starts with one action, advances automatically and resumes the same unfinished routine.
- A note-reading check respects clef, octave, key signature and accidentals.
- Held notes cannot answer repeated prompts; genuine new attacks still count.
- All 12 roots update notation and MIDI targets correctly; results stay separate by key.
- Wrong timed notes do not shift later targets; missing notes and extras remain visible.
- Pedal sustain stays separate from held keys and unverified technique claims.
- Hints, corrections and skips never become independent first-try successes.
- Lost Bluetooth links, duplicate inputs and save failure cannot corrupt the record.
- Keyboard navigation, accessible feedback and complete JSON restore work.

A reproducible assessment

Replaying stored targets and normalized events with the original evaluator version must reproduce the saved result.

Sources and document status

Platform references support feasibility. The lesson sequence, screens and evaluation thresholds are proposals.

1 MDN Web MIDI API

Secure contexts, permissions and availability limits.

2 MDN requestMIDIAccess

Access options, SysEx and failure states.

3 W3C Web MIDI API

Message-based access, ports and timestamped events; MIDI is separate from audio.

4 MIDI Association message summary

MIDI note, controller and other message categories.

5 ABRSM Piano syllabus 2025 and 2026

Reading, scales/arpeggios, application and aural work; no claimed alignment.

6 Apple Bluetooth MIDI setup

macOS MIDI Studio host connection and pairing flow.

7 Microsoft Windows MIDI Services releases

Release-specific BLE transport availability and prerequisites; verify browser compatibility.

The supplied indoor cycling Markdown and PDF informed this document's structure, device-readiness model, interruption behavior and portable-data approach. This is a new piano product proposal. References checked 9 October 2026.

Read together

The companion Markdown is the complete editable specification. This PDF presents the product direction and visual concepts.